

BENBECULA, United KingdomWMO: **030220**Lat: **57.481N**Lon: **7.363W**Elev: **6**StdP: **101.26**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.0	-0.2	-4.9	2.5	0.5	-3.4	2.8	1.6	20.2	6.4	18.2	6.6	2.2	70	0.803

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.1	20.1	15.8	18.2	14.6	17.0	14.2	16.4	18.9	15.5	17.6	14.8	16.6	5.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	10.8	17.2	14.8	10.5	16.6	14.0	10.0	15.6	45.9	18.6	43.2	17.3	41.2	16.8	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.4	15.0	13.4	-4.4	23.8	2.3	2.0	-6.0	25.2	-7.4	26.3	-8.7	27.4	-10.3	28.8		
(5)			-4.6	18.1	2.1	1.7	-6.1	19.3	-7.3	20.3	-8.5	21.2	-10.0	22.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	5.5	5.1	5.8	7.8	9.9	12.3	13.9	13.8	12.5	10.1	7.4	5.8
	DBStd	3.88	2.24	2.25	2.21	2.31	2.42	1.89	1.60	1.37	1.73	2.13	2.55	2.95
	HDD10.0	751	139	138	131	71	31	2	0	0	2	25	82	131
	HDD18.3	3343	397	371	389	315	263	181	137	141	176	256	329	388
	CDD10.0	450	0	0	0	6	26	71	122	117	76	27	4	1
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	3	0	0	0	0	1	1	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.7	7.8	7.0	7.1	6.4	6.1	5.7	5.6	5.8	6.6	7.0	7.1	7.5
Precipitation	PrecAvg	1284	153	109	106	75	63	71	82	99	107	137	138	136
	PrecMax	1694	262	254	215	122	159	132	142	265	208	233	225	240
	PrecMin	940	65	46	20	31	5	22	35	42	32	52	91	30
	PrecStd	171	48	51	41	23	35	27	25	46	41	54	31	53
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	11.1	13.0	17.8	21.9	22.1	22.5	21.0	18.2	16.0	13.1	12.5
		MCWB	9.8	9.2	9.9	11.4	15.3	16.7	17.3	16.3	14.7	13.8	11.2	11.3
	2%	DB	10.1	10.0	10.8	14.0	17.9	19.1	20.0	18.1	16.8	14.5	12.1	11.2
		MCWB	9.4	9.3	8.8	10.3	12.8	15.2	16.3	15.3	14.4	12.6	10.7	10.1
	5%	DB	9.4	9.2	9.9	12.1	15.4	17.1	18.1	17.0	15.9	13.9	11.7	10.8
		MCWB	8.7	8.6	8.6	9.2	11.8	13.9	15.1	14.5	13.8	12.2	10.4	9.8
	10%	DB	9.0	8.8	9.1	11.1	13.9	15.8	16.9	16.1	15.1	13.1	11.0	10.0
		MCWB	8.3	8.2	8.1	8.8	11.0	13.1	14.6	13.9	13.3	11.4	9.7	9.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	10.0	10.2	11.9	15.6	17.1	17.8	17.1	15.6	14.2	11.9	11.8
		MCDB	10.5	10.2	11.2	15.8	19.8	20.9	21.3	19.4	17.2	15.6	12.5	12.6
	2%	WB	9.5	9.3	9.4	10.7	13.8	15.4	16.7	15.7	15.0	13.2	11.3	10.7
		MCDB	9.7	9.5	10.0	13.0	17.3	18.6	19.2	17.5	16.2	14.4	12.1	11.3
	5%	WB	8.8	8.7	8.7	10.1	12.2	14.2	15.6	15.1	14.3	12.6	10.6	9.9
		MCDB	9.2	9.0	9.3	11.5	14.8	16.7	17.6	16.7	15.5	13.7	11.4	10.4
	10%	WB	8.1	7.7	8.1	9.4	11.2	13.3	14.8	14.5	13.7	11.9	9.9	9.1
		MCDB	8.7	8.2	8.9	10.8	13.3	15.2	16.5	15.9	14.9	13.0	10.7	9.7
Mean Daily Temperature Range	5% DB	MDBR	3.9	4.4	4.8	5.6	6.3	5.6	5.1	4.8	4.7	4.6	4.3	4.3
		MCDBR	3.7	3.7	5.3	7.3	9.3	8.7	7.9	6.2	5.5	5.2	4.1	4.4
		MCWBR	4.0	3.7	4.3	4.7	5.5	5.4	4.8	4.0	4.1	4.3	3.8	4.9
	5% WB	MCDBR	3.7	3.3	4.3	5.7	8.2	7.8	6.8	5.3	4.8	4.5	3.9	4.2
		MCWBR	4.2	3.7	4.0	4.2	5.4	5.2	4.5	3.8	4.1	4.3	4.1	4.8
Clear-Sky Solar Irradiance	taub	0.354	0.361	0.376	0.386	0.388	0.376	0.379	0.370	0.372	0.354	0.338	0.330	
	taud	2.124	2.232	2.283	2.313	2.366	2.445	2.449	2.479	2.430	2.355	2.199	2.108	
	Ebn at Noon	460	660	766	824	848	865	853	837	771	668	483	373	
	Edh at Noon	58	79	97	110	111	104	101	92	84	69	53	44	
All-Sky Solar Radiation	RadAvg	0.37	0.96	2.13	3.70	4.90	4.99	4.56	3.62	2.37	1.23	0.49	0.25	
	RadStd	0.04	0.09	0.16	0.26	0.37	0.50	0.32	0.23	0.11	0.12	0.04	0.03	